Test and Evaluation

Automated integration tests ran against actual MySQL 8.4.7 and an HTTP Node.js server. These results evaluate technical behavior; they are not participant feedback or a substitute for browser usability testing. The same sixteen tests also passed after creating a fresh database. Detailed executable steps and assertions are in tests/system.test.js.

Test ID	Requirement or scenario	Actual result	Evidence
T01	Registration and login	PASS	production-tests.txt
T02	Valid empty and invalid search	PASS	production-tests.txt
T03	Authentication and CSRF rejection	PASS	production-tests.txt
T04	Server fare and complete persistence	PASS	production-tests.txt
T05	Customer ownership	PASS	production-tests.txt
T06	Concurrent same-seat requests	PASS	production-tests.txt
T07	Multi-passenger rollback	PASS	production-tests.txt
T08	Cancellation and inventory release	PASS	production-tests.txt
T09	Malformed passenger and seat	PASS	production-tests.txt
T10	Administrator authorization	PASS	production-tests.txt
T11	Immutable booked fare	PASS	production-tests.txt
T12	Analytics reconciliation	PASS	production-tests.txt
T13	Airport and aircraft maintenance	PASS	production-tests.txt
T14	Full capacity and schedule cancellation	PASS	production-tests.txt
T15	Logout revocation	PASS	production-tests.txt
T16	MySQL FK and check constraints	PASS	production-tests.txt

T17 separately passed server-restart persistence; its saved output is evidence/persistence-check.txt. Tests use fictional accounts and remove their own test bookings afterward.

Critical Integrity Test Procedures

Test	Steps and expected result	Observed result
T06 concurrency	Two authenticated users submit the same free seat concurrently. Expect one HTTP 201, one HTTP 409 and one active allocation.	Passed: one accepted request and one conflict; count equals one.
T07 rollback	Submit two passengers: first requests a free seat; second requests an occupied seat. Expect failure with unchanged booking and passenger counts.	Passed: no partial booking or passenger records.
T08 cancellation	Cancel an owned confirmed booking. Retrieve status, inspect payment, and query seat availability.	Passed: cancelled, voided and seat available.
T11 fare change	Save booking totals, edit departure fare, read totals again, restore fare.	Passed: existing totals unchanged.
T14 full flight	Create a future departure; fill all 24 seats in groups of four; try another allocation; cancel departure.	Passed: extra booking rejected; all active seats released and bookings cancelled.
T17 restart	Create booking and session, stop Node.js, start it again, retrieve and cancel with saved session.	PASS: booking, passenger, price and session survived restart; cancellation then succeeded.

Defects and retests

A production test run failed because a database process in a different sandbox held the data directory and was unreachable. The preview context was stopped and the same suite was rerun in a shared execution context; all sixteen tests passed. This was not counted as a successful run. The initial test for rejected database writes was reviewed to ensure database connectivity had already been proven by the preceding tests.

Owner walkthrough and remaining checks

The owner completed a guided browser walkthrough on October 10, 2026. Screenshots support booking, administrator access, fare edits, analytics and printing. Seat release, login retrieval, narrow-window layout and keyboard navigation were reported by the owner. Evidence follows. Actual mobile-device and assistive-technology checks remain unverified.

Three User Evaluation Records

Three participant sessions are reported by the project owner and identified as P01, P02 and P03. The task set below covers search, booking, retrieval and cancellation. P03 also reviewed the administrator fare-change workflow. Fictional data was used; the owner's guided walkthrough is recorded separately.

Task	Record
Search a flight	Route/date chosen; completion; confusion; elapsed time if measured
Create a reservation	Passenger entry; seat selection; errors; assistance needed
Retrieve confirmation	Reference retrieval; clarity of fare/status; completion
Cancel reservation	Completion; understanding of released seats; error messages
Administrator review by at least one user	Create or edit a schedule; inspect reservation; role clarity

Ratings and feedback

The supplied participant ratings cover ease of use, clarity, reliability and overall satisfaction on a 1–5 scale. The final table incorporates the category scores supplied in the revised PDF. Comments are paraphrased from the owner; elapsed times and direct quotations were not supplied.

Participant	Session status	Actual ratings and comments
P01	Completed per owner; dates under review	Category ratings, feedback and admin task on final page.
P02	Completed per owner; dates under review	Category ratings, feedback and admin task on final page.
P03	Completed per owner; dates under review	Category ratings, feedback and admin task on final page.

The owner corrected the session order: P01 on October 2, P02 on October 5 and P03 on October 11, 2026. The wording was reportedly fixed on P01's test date and retested by P02 on October 5. Reconciliation with the recorded development timeline and identification of tested versions remain with the owner.

Evaluation Against Success Criteria

Criterion	Current evidence	Status
Correct fare calculation	T04 and T11	Verified by automated tests
Persistent bookings	MySQL integration plus restart check output	Verified by saved T17 PASS output
Duplicate-seat prevention	T06 plus database unique constraint	Verified
Atomic multi-passenger booking	T07 unchanged row counts after conflict	Verified
Cancellation consistency	T08 and T14	Verified
Role and ownership separation	T03, T05, T10 and T15	Verified
Fresh environment setup	clean-setup.txt	Verified on new database
Guided customer/admin browser flows	Owner walkthrough and screenshots on following pages	Observed/reported; limited to listed scenarios
Three real users complete tasks	Participant forms and actual feedback	Owner reports completion; dates and records incomplete

How to close the evaluation

For each real issue, add a defect ID, reproduction steps, expected and actual behavior, severity, correction and retest evidence. Update the implementation change log and affected requirements when a fix changes behavior. Do not replace earlier evidence silently. If the release changes, create a new commit and record the tested commit hash. Summarize remaining limitations and compare participant task completion with the original success criteria before submission.

Owner Walkthrough Results

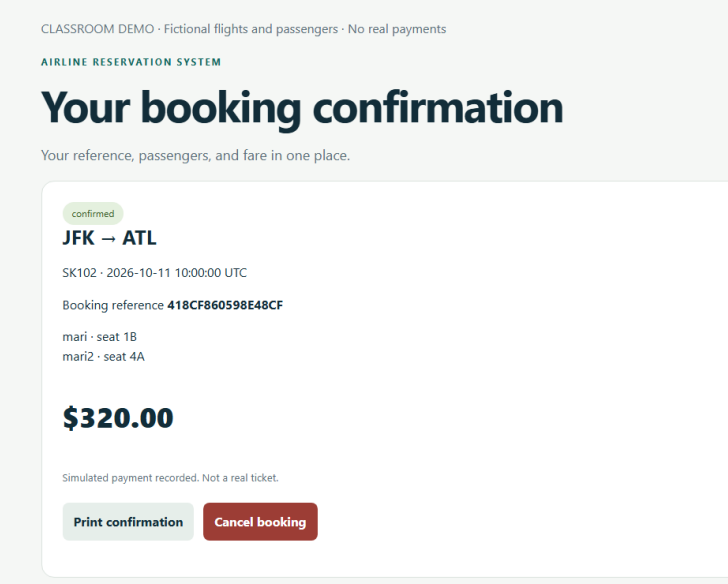
JahMari Harrigan performed this guided owner walkthrough on October 10, 2026 using the local application. Browser version and device model were not collected. No elapsed times or numerical ratings were collected. This is separate from the three required user evaluation sessions.

Check and expected behavior	Observed result	Evidence
M01 Book two seats at $160 each	Confirmed two passengers; saved total $320. PASS.	owner-booking.png
M02 Booking reduces seats; cancel releases them	Owner reported 22/24 after booking, 24/24 after cancellation. PASS by report.	Conversation; no cancellation screenshot
M03 Retrieve after logout/login	Owner reported booking remained retrievable. PASS by report.	Conversation; restart covered separately by T17
M04 Customer denied admin data; admin allowed	Denial screenshot and subsequent operations overview. PASS for observed pages.	owner-admin-denied.png; conversation
M05 Change SK102 fare to $175	Search showed the new $175 fare. PASS.	owner-fare-edit.png
M06 Analytics after one cancelled and one confirmed booking	2 reservations; 50% cancelled; $350; JFK-ATL 0.3%; zero integrity mismatches. PASS.	owner-analytics.png
M07 Print retains details and hides search	CLI fix plus final preview showed details on one page with search hidden. PASS.	cli-print-after.png
M08 Narrow window and keyboard Tab/Enter	Owner reported both worked. PASS by report; not a full accessibility audit.	Conversation

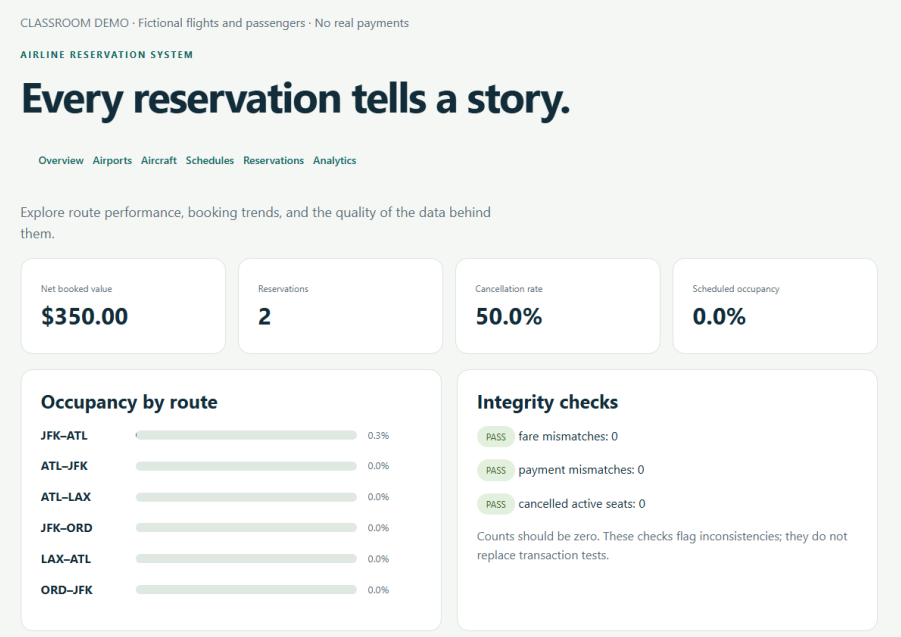
No new automated suite was run during this walkthrough or report update. Existing automated logs remain unchanged. A 24-seat result initially viewed was for the reverse route, not evidence of an inventory defect.

Booking and Analytics Evidence

Initial two-passenger confirmation at $160 per passenger. The later fare edit applied to new bookings.

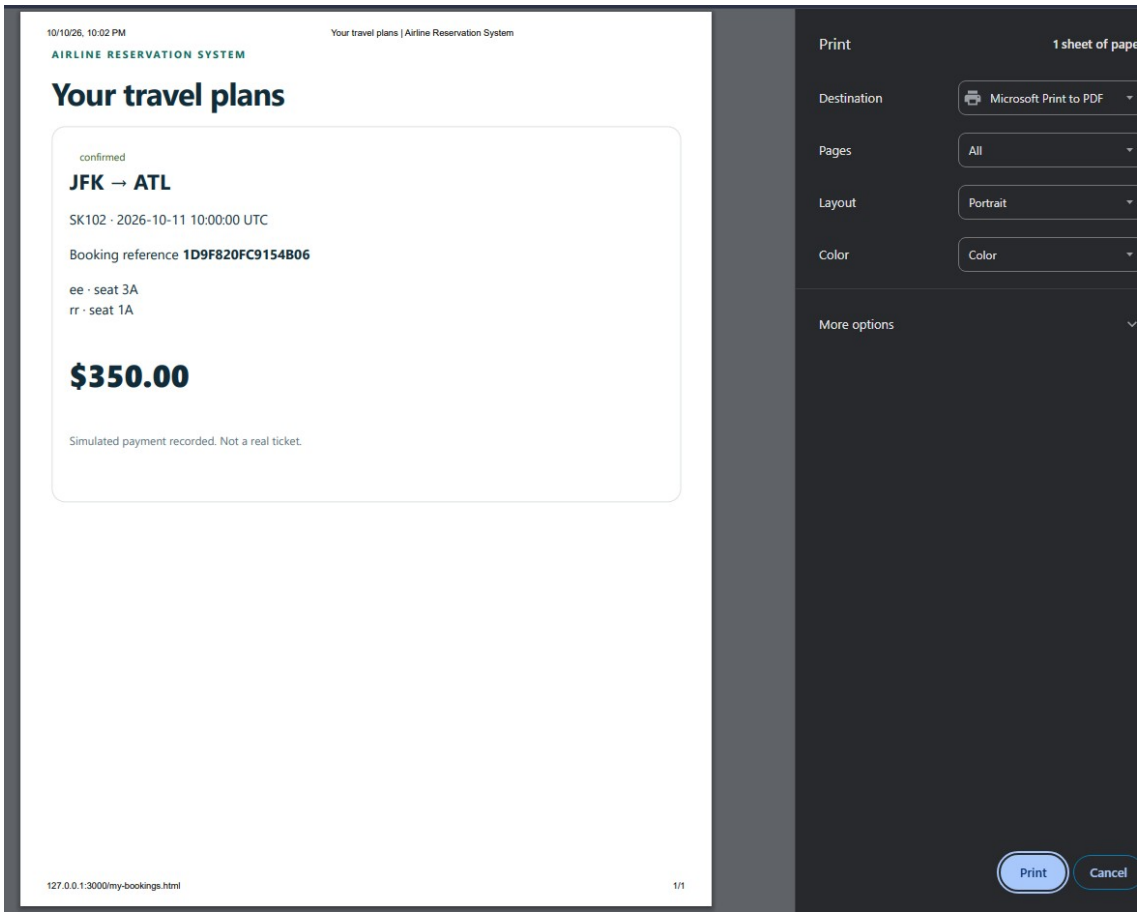


After cancellation and a new two-passenger booking at $175, the dashboard reconciled to $350 and 50% cancellations. Overall 0.0% is rounding of 2 occupied seats out of 4,320 scheduled seats; JFK-ATL is 2/720, displayed as 0.3%.



Print Defect and CLI Retest

PRINT-01: My Bookings print preview retained the lookup form and Show all link. The CLI added #lookup to the existing print-only hiding rule. Five source checks passed. The owner refreshed the page and supplied the preview below; lookup controls are absent and all booking details remain visible. No physical print or saved PDF was tested.



Prompt, changes, environment errors and static results: evidence/cli-print-review.md. Browser verification was performed by the owner and reviewed in desktop Codex after the CLI session. This does not retroactively change the earlier desktop development into CLI work.

Participant Feedback and Reported Retests

The owner reports three completed sessions. P01 reported unclear navigation on October 2; wording was clarified that day. P02 confirmed the improvement and tested other features on October 5. P03 tested on October 11, including administration. Ratings and outcomes below were supplied by the owner.

Participant and completion date	Browser and device	Rating supplied	Feedback supplied by owner
P01 October 2, 2026	Google Chrome Laptop	Ease of use: 3 Clarity: 2 Reliability: 3 Overall satisfaction: 3	One-word navigation tabs were too vague to explain their destination pages. Tasks worked; wording was clarified the same day, per owner.
P02 October 5, 2026	Google Chrome Laptop	Ease of use: 5 Clarity: 5 Reliability: 5 Overall satisfaction: 5	Successful run. Confirmed the navigation improvement and tested the other features on October 5, per owner.
P03 October 11, 2026	Opera Laptop	Ease of use: 5 Clarity: 5 Reliability: 5 Overall satisfaction: 5	Successful run. Admin task: changed SK102 JFK–ATL fare from $160 to $175 and checked the search result showed $175. Owner-reported participant outcome.

All four rating categories are now supplied. Tested version for P01, P02 and P03: not yet identified. Current workspace baseline is d019a03 with later uncommitted print and documentation changes; this is not proof of which build any participant tested. Browser versions and laptop operating systems were not supplied.

Issues and follow-up

USER-01: navigation and page context. The owner supplied “Flight” and “Find a flight” as examples when describing earlier wording changes, along with the notice “CLASSROOM DEMO · Fictional flights and passengers · No real payments” and “Built to be understood.” followed by “A systems analysis and data analytics project using fictional flights, transparent fares, and simulated payments.” These are recorded as owner-reported improvements, not requests for new interface changes. The owner reports Participant 2 confirmed improved clarity and attributes the later 5/5 ratings to it.

Current source navigation reads “Find a flight”; the demo notice and descriptive homepage text are present. The temporary edit to “Flight” was reverted. Earlier participant-tested wording chronology remains unconfirmed. No booking-saving defect is attributed to P01.

Remaining documentation: identify the actual tested release or commit for each participant and reconcile the October 2 and 5 sessions with the recorded project development timeline. The owner will resolve this separately.